CKA Exam Study Sheet: Kubernetes Troubleshooting Guide

A targeted cheatsheet for diagnosing and resolving failures across Master Nodes, Worker Nodes, and Kubernetes Resources during the CKA exam.

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1. Master Node (Control Plane) Troubleshooting

Control plane components usually run as **Static Pods** managed directly by `kubelet` via manifests in `/etc/kubernetes/manifests/`.

Quick Diagnostic Steps

```shell
# 1. Check control plane pod statuses
kubectl get pods -n kube-system

# 2. If API server is down, use crictl or docker directly on the master node
sudo crictl ps -a
sudo crictl logs <container-id>

# 3. Check kubelet status on master (manages static pods)
sudo systemctl status kubelet
sudo journalctl -u kubelet -e --no-pager
```

Common Failures & Fixes

Component	Common Symptoms	Root Cause / Check First	Common Solution
`kube-apiserver`	`The connection to the server ... was refused Error from server (InternalError)`	Syntax error in `/etc/kubernetes/manifests/kube-apiserver.yaml`, expired certificates, or `etcd` down.	Fix YAML typos in manifest; renew certs via `kubeadm certs renew all`.
`etcd`	`apiserver` crashing, `etcdserver: request timed out`	Disk full/slowness, expired etcd certs, corrupted data dir (`/var/lib/etcd`).	Check storage (`df -h`), check logs (`crictl logs <etcd-container>`). Restore from snapshot if corrupted.

Component	Common Symptoms	Root Cause / Check First	Common Solution
`kube-scheduler`	Pods remain in `Pending` state with no node assignment.	Static pod manifest broken or missing; controller crashed.	Inspect `/etc/kubernetes/manifests/kube-scheduler.yaml` logs.
`kube-controller-manager`	Deployments don't scale, PVCs stuck in `Pending`, node states not updating.	Manifest syntax error, missing service account keys or certificates.	Inspect `/etc/kubernetes/manifests/kube-controller-manager.yaml`.

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2. Worker Node Troubleshooting

Worker node issues usually cause nodes to report `NotReady` or fail to run workload pods.

Quick Diagnostic Workflow

```shell
# Step 1: Check node status and events
kubectl describe node <node-name>

# Step 2: SSH into the worker node
ssh <node-name>

# Step 3: Check system services
sudo systemctl status kubelet
sudo systemctl status containerd   # or cri-o / docker

# Step 4: Inspect system logs
sudo journalctl -u kubelet -e --no-pager
sudo journalctl -u containerd -e --no-pager
```

Common Worker Node Failures

Scenario A: Node in `NotReady` State

- **Cause 1: Swap enabled**
  - *Fix:* `sudo swapoff -a` and comment out swap entries in `/etc/fstab`.
- **Cause 2: Container Runtime down (`containerd`)**
  - *Fix:* `sudo systemctl restart containerd && sudo systemctl restart kubelet`
- **Cause 3: Network CNI plugin missing or crashed**

    - *Fix:* Check pods in `kube-system` (e.g., Flannel, Calico, Weave). Ensure CNI configuration exists in `/etc/cni/net.d/`.
- **Cause 4: Disk space exhausted**
    - *Fix:* Clear unused images/containers: `sudo crictl rmi --prune`.

Scenario B: Worker Node Cannot Join Cluster

- **Cause:** Expired join token or incorrect discovery hash.
- **Fix:** Generate a new join command on the master node:

```shell
kubeadm token create --print-join-command
```

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3. Resource & Workload Troubleshooting

Diagnostic Decision Tree

```
Pod Issue
 ├── State: Pending
 │    ├── Describe Pod: "0/N nodes available" -> Taints / Tolerations / Affinities
 │    ├── Check limits: Insufficient Memory/CPU on nodes
 │    └── PVC stuck in Pending -> StorageClass/PV binding issue
 │
 ├── State: CrashLoopBackOff / Error
 │    ├── Check application logs: kubectl logs <pod> [--previous]
 │    ├── Command/args misconfigured in pod spec
 │    └── Liveness probe failing
 │
 └── State: ImagePullBackOff / ErrImagePull
      ├── Typo in image name or tag
      └── Private registry missing Secret (imagePullSecrets)
```

Essential Resource Troubleshooting Commands

```shell
# --- Pods & Application Logs ---
kubectl describe pod <pod-name>
kubectl logs <pod-name> -c <container-name> --previous
kubectl exec -it <pod-name> -- /bin/sh

# --- Events & Debugging ---
```

```
kubectl get events --sort-by='.metadata.creationTimestamp' -n <namespace>
kubectl run debug-pod --rm -i --tty --image=busybox -- nslookup <service-name>

# --- Metrics & Resources ---
kubectl top nodes
kubectl top pods --all-namespaces

# --- Network & Services ---
kubectl get endpoints <service-name>
# If Endpoints are missing/empty: check Service selector labels match Pod labels!
```

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4. Key CKA Troubleshooting Checklist

1. **Systemd Services First:** When a node is `NotReady`, always check `systemctl status kubelet` and container runtime before anything else.
2. **Read the Logs End-to-End:** Use `journalctl -u kubelet -f` or `kubectl logs` with `--previous` to see what failed immediately before a crash.
3. **Validate Certificates:** Check expired certificates with:

Shell
```
kubeadm certs check-expiration
```

4. **Service Selectors:** If a Service cannot reach a Pod, check if `kubectl get ep <service-name>` lists IP addresses. If empty, fix label selectors in the Service definition.